AP PHYSICS 1 — IA-1 COMPLETE COURSE (Concept + Questions, From Scratch)

Learn every concept AND solve every level of question you have, then go harder.

HOW TO USE:

- Each block: READ the concept boxes first (understand, don't memorise). Then SOLVE the questions in order.
- Questions marked [HIS] are from your own WhatsApp worksheets / contests / Slack. Do these exactly.
- Questions marked [HARDER] are the same concept one step up — what he could ask beyond your bank.
- Check answers only AFTER you write your own.

BLOCK 1 — MEASUREMENT & UNITS (L1)

CONCEPT 1.1 — What measurement is

Measurement = a **number + a unit** that tells how much of something you have.

Example: length = 25.60 cm. The number alone (25.60) means nothing without "cm".

Why engineering needs it: quality control, comparing across teams, manufacturing tolerances, safety limits, scientific communication. Two teams using different units → disaster (see CONCEPT 1.4).

CONCEPT 1.2 — Fundamental vs derived quantities

- **Fundamental** = defined on its own, not from others. The 7 SI ones: mass (kg), length (m), time (s), temperature (K), electric current (A), amount of substance (mol), luminous intensity (cd).
- **Derived** = built from fundamentals. Speed = length/time. Force = mass × acceleration.

CONCEPT 1.3 — Dimensions

Write every quantity using **M** (mass), **L** (length), **T** (time).

- Velocity = L/T = $\mathbf{LT^{-1}}$
- Acceleration = velocity/time = $\mathbf{LT^{-2}}$
- Force = mass × acceleration = $\mathbf{MLT^{-2}}$
- Work = force × distance = $\mathbf{ML^{2}T^{-2}}$
- Pressure = force/area = $\mathbf{ML^{-1}T^{-2}}$

Rule of the game: in any correct equation, every term you add must have the SAME dimensions. That is a **dimensional check**.

CONCEPT 1.4 — Scientific notation & order of magnitude

- Scientific notation: write as $a \times 10^n$ where $1 \le a < 10$.

$0.000456 = 4.56\times10^{-4}$. $7{,}850{,}000 = 7.85\times10^{6}$.

- Order of magnitude: look at the leading digit of a. If a ≥ 5, round the power UP by 1; if < 5, keep it.

$7.85\times10^{6} \rightarrow \mathbf{10^{7}}$. $4.56\times10^{-4} \rightarrow \mathbf{10^{-4}}$. $8.2\times10^{-9} \rightarrow \mathbf{10^{-8}}$.

CONCEPT 1.5 — Errors

Three families (his favourite MCQ):

- **Random error** = unpredictable up/down scatter (parallax, vibration). Fix: average many readings.
- **Systematic error** = one-way consistent bias (zero error, wrong calibration). Averaging does NOT remove it.
- **Gross error** = human blunder (misread, wrong button). Fix: re-measure carefully.

Three numbers you can always compute:

- **Absolute error** Δx = |measured − true| (has units)
- **Relative error** = Δx / true (no units)
- **Percentage error** = relative × 100

CONCEPT 1.6 — Error propagation (how errors travel through a formula)

- **Add/subtract** quantities → their **absolute** errors add.
- **Multiply/divide** quantities → their **relative** errors add.
- Raise to power n → the relative error gets **multiplied by n**.
- Constants (like π) carry **no error**.

CONCEPT 1.7 — Significant figures

Count the meaningful digits. Leading zeros do NOT count.

0.00456 → **3** significant figures (4, 5, 6).

HIS QUESTIONS (do these exactly)

[HIS Q1] Why is measurement important in engineering? (write 4–5 reasons)

[HIS Q2] Name any 5 fundamental SI quantities and their units.

[HIS Q3] Write dimensional formulas of velocity, acceleration, force, work, pressure.

[HIS Q4] Write in scientific notation + give order of magnitude: 0.000456; 7,850,000; 299,792,458; 0.0000000082.

[HIS Q5] A rod is measured 25.60 cm; true value is 25.30 cm. Find absolute, relative, percentage error.

[HIS Q6] Disc radius r = (10.0 ± 0.2) cm. Find the % error in area $A = \pi r^2$.

[HIS Q7] A = (50.0 ± 0.5), B = (20.0 ± 0.2). Find the max absolute and relative error in Z = A/B.

[HIS Q8] Check dimensionally whether $s = ut + \frac{1}{2}at^2$ is correct.

[HIS Q9] Explain the Mars Climate Orbiter failure: what units were mixed, what happened, and list lessons a company should enforce.

[HIS Q10] Cube side = (5.00 ± 0.02) cm. Find (a) % error in volume (b) max absolute error in volume.

[HIS Q11] Same story question as Q9 in another form — retell what happened and the lessons.

[HIS CONTEST] What type of error comes from unpredictable variations in measurements? (answer: random error)

[HIS CONTEST] What is the order of magnitude of 0.00000487 and 564000000?

ANSWERS

Q1 → Measurement = number + unit. Engineering needs: accuracy/quality control, comparison across teams, manufacturing consistency/tolerances, scientific communication, safety limits. Missing a standard unit system caused a real spacecraft loss.

Q2 → mass kg, length m, time s, temperature K, current A (any 5 of the 7).

Q3 → LT^{-1}, LT^{-2}, MLT^{-2}, ML^2T^{-2}, $ML^{-1}T^{-2}$.

Q4 → 4.56×10^{-4} (10^{-4}); 7.85×10^{6} (10^{7}); 2.998×10^{8} (10^{8}); 8.2×10^{-9} (10^{-8}).

Q5 → ΔL = 0.30 cm; relative = 0.30/25.30 = 0.0119; % = 1.19%.

Q6 → ΔA/A = 2 × (0.2/10.0) = 0.04 → **4%**. (power rule: the square multiplies the relative error by 2; π has no error.)

Q7 → relative = 0.5/50 + 0.2/20 = 0.01 + 0.01 = 0.02 = 2%. Z = 2.5, so ΔZ = 0.02 × 2.5 = 0.05 → **Z = 2.5 ± 0.05**.

Q8 → [s] = L. [ut] = $LT^{-1}\cdot T$ = L. [at^2] = $LT^{-2}\cdot T^2$ = L. All terms L → dimensionally correct.

Q9/Q11 → One team sent thrust in pound-force·seconds (Imperial), the navigation team expected newton·seconds (SI). Same equation, two unit systems → model wrong by ~4.45× → craft entered the atmosphere too low and was lost (~$125M, 1999). Lessons: enforce SI-only pipelines, unit-tag every variable, automated unit-checking software, dimensional analysis as a sanity check, standardised interfaces between teams.

Q10 → $V = s^3$. % error = 3 × (0.02/5.00) = 0.012 = **1.2%**. $\Delta V = 3s^2\Delta s = 3\times25\times0.02$ = **1.5 cm^3**.

Contest → random error. 0.00000487 → 4.87×10^{-6} → **10^{-5}** (4.87 ≥ √10≈3.16, round up). 564000000 → 5.64×10^{8} → **10^{9}**.

[HARDER] — go beyond your bank

[H1] A length is measured as (12.5 ± 0.1) cm and time as (2.00 ± 0.02) s. Speed = L/T. Find the speed and its max percentage error. (answer ≈ 6.25 m/s, 2%)

[H2] Explain why averaging readings reduces random error but NOT systematic error. (answer: random errors scatter both ways and cancel in the mean; systematic errors shift every reading the same way, so the mean keeps the bias.)

[H3] A student measures g = 9.81, 9.83, 9.79, 9.82, 9.80 m/s^2. True value 9.80. Comment on random vs systematic error present. (answer: scatter around the mean = small random error; the mean 9.81 vs true 9.80 = tiny systematic bias.)

BLOCK 2 — VECTORS & CALCULUS TOOLS (L2)

CONCEPT 2.1 — Scalar vs vector

- **Scalar**: magnitude only (mass, time, speed, energy).
- **Vector**: magnitude + direction (displacement, velocity, acceleration, force, momentum).

Represent a vector with an arrow, or in components **A = (Ax, Ay)**, magnitude $|A| = \sqrt{(Ax^2 + Ay^2)}$, direction $\theta = \tan^{-1}(Ay/Ax)$.

CONCEPT 2.2 — Vector addition

- **Triangle law**: place vectors head-to-tail; the resultant closes the triangle.

- **Parallelogram law**: place tail-to-tail, complete the parallelogram; the diagonal is the sum.
- Magnitude of A + B = $\sqrt{(A^2 + B^2 + 2AB\cos\theta)}$ where θ is the angle between them.
- Adding components: (A+B)x = Ax + Bx, same for y. This is the method you will use for every problem.

CONCEPT 2.3 — Multiplying a vector by a scalar

kA: magnitude scales by |k|. If k is **negative, the direction reverses**. (His MCQ: vector × negative scalar → direction is reversed.)

CONCEPT 2.4 — Two products (his 2-mark comparison)

- **Dot product** A·B = AB cosθ. Result is a **scalar**. Perpendicular → cos90° = 0, so A·B = 0. Commutative (A·B = B·A).
- **Cross product** A×B = AB sinθ n̂. Result is a **vector**, perpendicular to both A and B, direction by the **right-hand rule**. Parallel → sin0° = 0, so A×B = 0. NOT commutative (A×B = −B×A).

CONCEPT 2.5 — Unit vector and resolution

Unit vector $\hat{A} = A/|A|$ (magnitude 1, keeps direction).

Any vector resolved: A = Ax î + Ay ĵ. To resolve a vector of magnitude A at angle θ to the x-axis: Ax = A cosθ, Ay = A sinθ.

CONCEPT 2.6 — Differentiation (how fast a quantity changes)

The derivative gives the instantaneous rate of change. On a graph, it is the **slope of the tangent**.

- Power rule: $d/dx\ (x^n) = n\ x^{n-1}$. Multiply by the power, then drop the power by 1. Constants die.
- $d/dx\ (4x^2 - 3x + 7) = 8x - 3$.
- In physics: position → velocity (v = dx/dt) → acceleration (a = dv/dt).

CONCEPT 2.7 — Integration (the reverse — accumulating)

- Power rule: $\int x^n\,dx = x^{n+1}/(n+1) + C$. Raise the power by 1, divide by the new power. Always add **+ C**.
- $\int (3x^2 + 2x)\,dx = x^3 + x^2 + C$.
- On a graph, the integral is the **area under the curve**.
- In physics: acceleration → velocity (v = ∫ a dt) → position (x = ∫ v dt). The constants are fixed by the starting (initial) conditions.

HIS QUESTIONS

[HIS Q18a] Find dy/dx of $y = 4x^2 - 3x + 7$.

[HIS Q18b] Integrate $\int (3x^2 + 2x)\,dx$.

[HIS CONTEST] What happens to a vector when multiplied by a negative scalar? (answer: magnitude scales, direction reverses)

[HIS CONTEST] According to the triangle law, vectors are arranged how? (answer: head-to-tail)

[HIS CONTEST] Dot product of two perpendicular vectors is? (answer: 0)

[HIS CONTEST] Which determines the direction of a cross product? (answer: the right-hand rule)

[HIS CONTEST] Differentiate dot product from cross product (at least two differences each).

ANSWERS

Q18a → $8x - 3$.

Q18b → $x^3 + x^2 + C$.

Q17 → Dot: scalar result, $A \cdot B = AB\cos\theta$, commutative, zero when perpendicular. Cross: vector result, $A \times B = AB\sin\theta\,\hat{n}$, right-hand rule direction, anti-commutative, zero when parallel.

[HARDER]

[H4] If $A = 3\hat{i} + 4\hat{j}$ and $B = 2\hat{i} - \hat{j}$, find $A \cdot B$ and $|A \times B|$. (answer: $A \cdot B = 6 - 4 = 2$; $|A \times B| = |3(-1) - 4(2)| = 11$)

[H5] d/dx of $y = 5x^3 - 2x + 1$, then integrate your answer back. (answer: $15x^2 - 2$; back = $5x^3 - 2x + C$)

[H6] A vector has magnitude 10 at 30° to the x-axis. Resolve it. (answer: $8.66\hat{i} + 5\hat{j}$)

BLOCK 3 — KINEMATICS (L3–L4)

CONCEPT 3.1 — The basic quantities

- **Distance** = path length travelled (scalar). **Displacement** = change in position (vector, straight line from start to finish).
- **Speed** = distance/time (scalar). **Velocity** = displacement/time (vector).
- **Acceleration** = rate of change of velocity (vector). **Jerk** = rate of change of acceleration. (His MCQ trap: "rate of change of velocity" = acceleration, NOT jerk.)

CONCEPT 3.2 — The three equations of motion (constant acceleration)

Only valid when a is constant:

1. $v = u + at$ (no distance involved)

2. $v^2 = u^2 + 2as$ (no time involved)

3. $s = ut + \frac{1}{2}at^2$ (no final velocity involved)

where u = initial velocity, v = final, a = acceleration, t = time, s = displacement.

Pick by what is MISSING from the question.

CONCEPT 3.3 — Graphs (the ladder)

- Position–time graph: **slope = velocity**.
- Velocity–time graph: **slope = acceleration**, **area = displacement**.
- Straight rising line = constant velocity. Curved steepening line = increasing velocity.

CONCEPT 3.4 — Relative motion

$v(A/B) = v_A - v_B$ (velocity of A relative to B). Pick a positive direction and write it down; then signs take care of themselves.

- River crossing: time to cross = width / boat-speed-perpendicular; drift = stream-speed × time.
- Rain and umbrella: the umbrella tilts at $\tan\theta = v_{rain} / v_{man}$.

CONCEPT 3.5 — Projectile motion

Split the launch velocity into components, then treat the two directions separately (horizontal = constant velocity, vertical = constant acceleration $-g$).

- $u_x = u\cos\theta$, $u_y = u\sin\theta$
- Time of flight $T = 2u_y/g$
- Max height $H = u_y^2/(2g)$
- Range $R = u^2 \sin 2\theta / g$. Range is maximum at **44–45°**; complementary angles (θ and $90-\theta$) give the same range.
- At any time: $v_x = u_x$ (constant), $v_y = u_y - gt$.

HIS QUESTIONS

[HIS Q4] Two cars on the same road: A at 20 m/s, B at 12 m/s, same direction. Velocity of A relative to B? If opposite directions?

[HIS Q19] Two position-time graphs — one a straight rising line, one curving upward. State how average and instantaneous velocity change in each.

[HIS Q20] A body starts from rest with $a = 2$ m/s². What is its velocity after 5 s?

[HIS Q24] A ball is kicked at 25 m/s, 37° to the horizontal. Find (1) horizontal and vertical components (2) time to return to the ground (3) max height. ($\sin 37 = 0.6$, $\cos 37 = 0.8$)

[HIS Q25] A drone's position is $x(t) = 3t^2 + 4t + 2$ and $y(t) = 2t^3 - 3t^2 + 5t$. Find velocity components and speed at $t = 2$ s; acceleration components and magnitude at $t = 2$ s; direction of motion.

[HIS CONTEST] In projectile motion the vertical acceleration is? (answer: $-g$)

ANSWERS

Q4 → Same direction: $20 - 12 =$ **8 m/s**. Opposite: $20 - (-12) =$ **32 m/s**.

Q19 → Straight line: velocity constant, average = instantaneous, both unchanging. Curving upward: velocity increases with time; instantaneous velocity (tangent slope) grows; average over an interval is less than the instantaneous at the end.

Q20 → $v = u + at = 0 + 2\times5 =$ **10 m/s**.

Q24 → $u_x = 25\times0.8 =$ **20 m/s**, $u_y = 25\times0.6 =$ **15 m/s**. $T = 2\times15/g$. With $g=10$ → **3.0 s** (with $g=9.8$ → 3.06 s). $H = 15^2/(2\times10) =$ **11.25 m** (with $g=9.8$ → 11.5 m).

Q25 → $v_x = 6t+4$ → **16**, $v_y = 6t^2-6t+5$ → **17** at $t=2$. Speed $= \sqrt{(16^2+17^2)} =$ **$\sqrt{545} \approx$ 23.3 m/s**. $a_x = 6$, $a_y = 12t-6$ → **18**. $|a| = \sqrt{(6^2+18^2)} =$ **$\sqrt{360} \approx$ 19.0 m/s²**. Direction $\theta = \tan^{-1}(17/16) \approx$ **46.7°** from +x.

[HARDER]

[H7] A ball is thrown horizontally at 15 m/s from a 45 m high cliff. Where does it land? (answer: $t = \sqrt{(2\times45/10)} = 3$ s; range $= 15\times3 = 45$ m)

[H8] A car goes 0→20 m/s in 5 s, then holds 20 m/s for 5 more. Using the velocity–time graph, find displacement, average velocity, average acceleration in the first 5 s. (answer: 100 m, 10 m/s, 4 m/s² — area under the graph)

[H9] A boat points straight across a 200 m river at 4 m/s while the current flows at 3 m/s. Crossing time and downstream drift. (answer: 50 s, 150 m)

BLOCK 4 — FORCES, NLM & FRICTION (L5–L6)

CONCEPT 4.1 — What a force is

A push or pull from one object on another. Types you will meet: weight (mg, down), normal force (N, perpendicular to the surface), tension (T, along a string), friction (f, opposes slipping), applied force, spring force.

CONCEPT 4.2 — Newton's three laws

1. **First law (inertia):** an object stays at rest or moves with constant velocity unless a net external force acts. This defines an **inertial frame**.

2. **Second law:** F_net = ma (net force = mass × acceleration). This is the workhorse.

3. **Third law:** every action has an equal and opposite reaction (the two forces act on DIFFERENT objects, so they never cancel in one free-body diagram).

CONCEPT 4.3 — Frames of reference and pseudo forces

- **Inertial frame:** not accelerating; Newton's laws hold directly.
- **Non-inertial (accelerating) frame:** to make the laws appear to hold, you must add a **pseudo force** = − (acceleration of the frame) × mass, opposite to the frame's acceleration. (Example: in an accelerating car you feel thrown backward — that "force" is the pseudo force.)

CONCEPT 4.4 — Free body diagrams (FBD) — his constant companion

Draw ONE object. On it, draw every force acting ON that object as an arrow:

- Box on a floor: N up, mg down, applied force, friction opposite to motion.
- Box on an incline: split mg into mg sinθ (down the slope) and mg cosθ (into the slope). N = mg cosθ. Friction acts up the slope if it is sliding down.

Always draw the FBD before writing any equation. He draws them in his notes and rewards them.

CONCEPT 4.5 — Friction

- **Static friction** f_s: opposes the tendency to move. Self-adjusting — it equals the applied force up to a maximum f_s,max = μ_s N. The object moves only when the applied force exceeds f_s,max.
- **Kinetic friction** f_k = μ_k N: constant, acts once sliding. Usually μ_k < μ_s.
- On an incline: the block slides if the down-slope component beats max static friction. Quick test: **slides if tanθ > μ_s**.

- Pulling UP at an angle reduces the normal force (N = mg − F sinθ). Pushing DOWN at an angle increases it (N = mg + F sinθ).

CONCEPT 4.6 — Whole-system method (for pulleys)

If two masses are tied by a light inextensible string over a frictionless pulley, they share one acceleration a.

a = (driving force − resisting force) / total mass.

Then find the tension from ONE of the bodies alone.

HIS QUESTIONS

[HIS F1] 5 kg, μ_s = 0.40. What is the maximum force you can apply without moving it?

[HIS F2] 10 kg pulled by 35 N, μ_k = 0.20. Find N, friction, net force, acceleration.

[HIS F3] 6 kg, μ_s = 0.50, μ_k = 0.30, applied force 20 N. Does it move? What is the actual friction?

[HIS F4] 8 kg, μ_s = 0.40, μ_k = 0.25, force slowly increased from 0. At what force does it start moving? Just after moving, what is friction and acceleration?

[HIS F5] 5 kg, F = 40 N applied at 30° above the horizontal, μ_k = 0.20. Find acceleration.

[HIS F6] 10 kg, F = 50 N applied at 30° below the horizontal (a push), μ_k = 0.25. Find acceleration.

[HIS F7] 10 kg on a 30° incline, μ_k = 0.20, sliding down. Find acceleration.

[HIS F8] 5 kg on a 37° incline, μ_s = 0.50. Does it stay or slide? (g = 10)

[HIS F9] 4 kg on a 30° incline, F = 30 N applied up the slope, μ_k = 0.20. Find acceleration.

[HIS F10] 6 kg pulled up a 30° incline by a 50 N rope, μ_k = 0.15. Find acceleration.

[HIS F11] 5 kg on a rough table (μ_k = 0.20) tied to a hanging 3 kg over a frictionless pulley. Find acceleration and tension.

ANSWERS (g = 9.8 unless stated)

F1 → f_s,max = 0.40 × 5 × 9.8 = **19.6 N**.

F2 → N = 98 N, f_k = 19.6 N, net = 15.4 N, a = **1.54 m/s²**.

F3 → f_s,max = 29.4 N > 20 N, so it does NOT move. Actual friction = **20 N** (self-adjusting, matches the applied force).

F4 → Starts at 0.40×8×9.8 = **31.36 N**. Just after: f_k = 0.25×8×9.8 = **19.6 N**, a = (31.36 − 19.6)/8 = **1.47 m/s²**.

F5 → F_x = 34.64 N, F_y = 20 N, N = 49 − 20 = 29 N, f_k = 5.8 N, a = (34.64 − 5.8)/5 = **5.77 m/s²**. (Pulling up reduces N, so a is larger.)

F6 → N = 98 + 25 = 123 N, f_k = 30.75 N, a = (43.30 − 30.75)/10 = **1.26 m/s²**. (Pushing down increases N, so a is smaller.)

F7 → a = g(sin30 − μ_k cos30) = 9.8(0.5 − 0.20×0.866) = **3.20 m/s²** down the slope.

F8 → tan37 = 0.75 > μ_s = 0.50 → **it slides**.

F9 → N = 4×9.8×cos30 = 33.95 N, f_k = 6.79 N, down-slope weight = 19.6 N. Net up = 30 − 19.6 − 6.79 = 3.61 N → a = **0.90 m/s² up the slope**.

F10 → N = 50.92 N, f_k = 7.64 N, down-slope weight = 29.4 N, a = (50 − 29.4 − 7.64)/6 = **2.16 m/s² up**.

F11 → a = (3×9.8 − 0.20×5×9.8)/(5+3) = (29.4 − 9.8)/8 = **2.45 m/s²**. T = 3(9.8 − 2.45) = **22.05 N**.

[HARDER]

[H10] A 15° incline, $\mu_k = 0.10$, 20 kg block. Does it slide? If yes, find a. (answer: $\tan 15 = 0.27 > 0.10 \rightarrow$ slides; $a = 9.8(\sin 15 - 0.10 \cos 15) = 9.8(0.259 - 0.097) = 1.59\ m/s^2$)

[H11] Three blocks (2 kg, 3 kg, 4 kg) tied on a frictionless table, pulled by 18 N. Find a and each tension. (answer: $a = 2\ m/s^2$; T between 4 and 3 = 4×2 = 8 N; T between 3 and 2 = 2×2 = 4 N)

[H12] A 10 kg block on a rough horizontal table ($\mu_k = 0.30$) is pulled by a rope at 20° above horizontal with 50 N. Find a. (answer: $N = 98 - 50 \sin 20 = 98 - 17.1 = 80.9$ N; $f_k = 24.27$ N; $a = (50 \cos 20 - 24.27)/10 = (46.98 - 24.27)/10 = 2.27\ m/s^2$)

BLOCK 5 — CIRCULAR MOTION (L7)

CONCEPT 5.1 — Angular description

An angle in **radians**: θ (radians) = arc length / radius = s/r. Full circle = 2π rad.

- **Angular velocity** $\omega = \theta/t = v/r = 2\pi f = 2\pi/T_{period}$.
- **Angular acceleration** $\alpha = \omega/t$ (the angular version of a).

Rotational kinematics mirror the linear ones: θ ↔ x, ω ↔ v, α ↔ a.

CONCEPT 5.2 — Radial and tangential directions

At any point on a circle:

- **Tangential** direction: along the tangent, where the linear speed lives. Tangential acceleration $a_t = r\alpha$ (changes the SPEED).
- **Radial (centripetal) direction**: toward the centre. Centripetal acceleration $a_c = v^2/r = r\omega^2$ (changes the DIRECTION, not the speed).
- Total acceleration $a = \sqrt{(a_t^2 + a_c^2)}$.

CONCEPT 5.3 — Centripetal vs centrifugal (his favourite explanation)

- **Centripetal** = the REAL inward net force (tension, friction, gravity) that constantly redirects the velocity so the circular path continues. Acts ON the body, in an inertial frame. Agent: a string, friction, gravity.
- **Centrifugal** = a **pseudo (fictitious) force** that appears only in the rotating (non-inertial) frame, directed radially OUTWARD. No physical agent — it represents the body's inertia resisting the inward turn. Example: the outward push you feel on a merry-go-round; clothes flung outward in a spinning dryer.
- Key point: centripetal does NO work, because it is always perpendicular to the motion ($W = Fd\cos 90° = 0$).

CONCEPT 5.4 — Torque

$\tau = rF\sin\theta$, where θ is the angle between the radius vector and the force. It is zero when the force points along the radius ($\sin 0° = 0$) — the same reason centripetal force does no work.

HIS QUESTIONS

[HIS C1] A car moves in a circle of radius 50 m at 10 m/s. Find ω and a_c.

[HIS C2] A wheel of radius 0.5 m spins at 120 rpm. Find ω and the linear speed v.

[HIS C3] Torque from a 100 N force at radius 0.5 m when the force is at 90°, 60°, 0° to the radius.

[HIS C4] 2 kg, r = 4 m, v = 8 m/s. Find centripetal acceleration and the centripetal force.

[HIS C5] A 1200 kg car goes around a bend of radius 40 m at 12 m/s. Find the required centripetal force.

[HIS C6] A 5 kg mass spins on a string of length 2 m at 6 m/s. What is the tension?

[HIS C7] Explain the difference between centripetal and centrifugal force, with an example of each.

[HIS C8] A 5 kg block on a rough turntable (radius 2 m, μ = 0.4). What is the maximum speed it can have without slipping?

[HIS C9] A car travels at 20 m/s around a bend of radius 100 m. Find a_c. What happens to a_c if the speed doubles?

ANSWERS

C1 → ω = 10/50 = **0.2 rad/s**; a_c = 10^2/50 = **2 m/s²**.

C2 → ω = 120 × 2π/60 = **4π ≈ 12.57 rad/s**; v = 0.5 × 4π = **2π ≈ 6.28 m/s**.

C3 → 90°: **50 N·m**. 60°: **43.3 N·m**. 0°: **0**.

C4 → a_c = 8^2/4 = **16 m/s²**; F = 2 × 16 = **32 N**.

C5 → F = 1200 × 144/40 = **4320 N**.

C6 → T = mv^2/r = 5×36/2 = **90 N**.

C7 → See CONCEPT 5.3. Centripetal = real inward force (string tension, friction, gravity) in an inertial frame. Centrifugal = pseudo outward force felt in the rotating frame, no physical agent.

C8 → μmg = mv^2/r → v = √(μgr) = √(0.4×9.8×2) = **2.8 m/s**.

C9 → a_c = 400/100 = **4 m/s²**. If v doubles, a_c quadruples (**16 m/s²**) because a_c depends on v^2.

[HARDER]

[H13] A satellite orbits at radius r with speed v. Derive the orbital speed if gravity provides the centripetal force. (answer: mv^2/r = GMm/r^2 → v = √(GM/r))

[H14] A wheel accelerates from ω = 2 to 8 rad/s in 4 s at radius 0.3 m. Find a_t, a_c at the end, and total a. (answer: α = 1.5; a_t = 0.45; a_c = 0.3×64 = 19.2; a ≈ 19.2 m/s² — dominated by the radial part)

[H15] Why does a car need a larger banking angle for a higher speed on the same curve? (answer: the horizontal component of the normal force must supply mv^2/r, which grows with v^2, so a larger angle is needed for the same radius.)

BLOCK 6 — WORK & ENERGY (L8)

CONCEPT 6.1 — Work

Work done by a constant force: **W = F d cosθ**, where θ is the angle between the force and the displacement.

- Positive work if the force has a component along the motion; negative if opposite (friction).
- **Zero work** when the force is perpendicular to the motion (cos90° = 0). Example: centripetal force on a circular path, normal force on a flat road, gravity at constant height.

CONCEPT 6.2 — Energy

- **Kinetic energy** KE = ½mv² (energy of motion).
- **Gravitational potential energy** PE = mgh (energy of position).
- **Spring potential energy** = ½kx².
- **Power** = work/time = Fv.

CONCEPT 6.3 — Work–energy theorem

W_net = ΔKE — the net work done on an object equals its change in kinetic energy. This often replaces the equations of motion when forces vary or when you only care about speeds.

CONCEPT 6.4 — Conservative vs non-conservative forces

- **Conservative** (gravity, spring force): the work done depends ONLY on the start and end points, not on the path taken. Over any closed loop the work is zero. Hence KE + PE = constant (mechanical energy conserved).
- **Non-conservative** (friction, air resistance): the work depends on the path — a longer path means more energy removed. Mechanical energy is NOT conserved; it turns into heat. W_friction = –f_k × d.

HIS QUESTIONS

[HIS E1] 5 kg moving at 4 m/s. What is its kinetic energy?

[HIS E2] An object has momentum 10 kg·m/s and mass 2 kg. Find its kinetic energy.

[HIS E3] 5 kg at a height of 10 m. What is its gravitational potential energy?

[HIS E4] An 80 N force acts over 8 m in the direction of the force. Find the work done.

[HIS E5] A 100 N force acts over 5 m at 60° to the displacement. Find the work done.

[HIS E6] A spring of spring constant 200 N/m is stretched from 0.1 m to 0.3 m. Find the work done by the spring.

[HIS E7] The same amount of work is done moving an object along two different paths between the same two points. What does this tell you about the force?

[HIS E8] A 4 kg object at rest has 200 J of net work done on it. Find its final speed.

[HIS E9] A 5 kg object moving at 4 m/s has 90 J of work done on it. Find its final speed.

[HIS E10] A 2 kg object falls freely from rest off a height of 20 m. Find its speed just before hitting the ground.

[HIS E11] A 5 kg object falls 8 m and loses 50 J to friction. Find its speed at the bottom.

[HIS E12] A 10 kg object falls 5 m and loses 100 J to friction. Find its speed at the bottom.

[HIS E13] A 4 kg block is pushed by 30 N through 10 m against 5 N of friction, from rest. Find: work by the push, work by friction, net work, final KE, final speed.

[HIS E14] KE = ½mv² is kept constant while the mass becomes 4m. What happens to the speed?

[HIS E15] A particle falls under gravity with friction present. Is total mechanical energy conserved between start and finish? Explain.

ANSWERS

E1 → ½×5×16 = **40 J**.

E2 → K = $p^2/2m$ = 100/4 = **25 J**.

E3 → 5×9.8×10 = **490 J**.

E4 → **640 J**.

E5 → 100×5×cos60 = **250 J**.

E6 → $-\frac{1}{2}k(x_2^2 - x_1^2)$ = −100(0.09 − 0.01) = **−8 J** (the spring does negative work when stretched further; the external agent does +8 J).

E7 → The force is **conservative** — work is path-independent, so over a closed loop it is zero.

E8 → 200 = ½×4×v^2 → v = **10 m/s**.

E9 → 90 = ½×5(v^2 − 16) → v = **7.21 m/s**.

E10 → v = √(2gh) = √(2×9.8×20) = **19.8 m/s**.

E11 → mgh − 50 = ½mv^2 → 392 − 50 = 2.5v^2 → v = **11.7 m/s**.

E12 → 490 − 100 = 5v^2 → v = **8.83 m/s**.

E13 → W_{push} = 300 J; $W_{friction}$ = −50 J; W_{net} = 250 J; KE_{final} = 250 J; v = √(2×250/4) = **11.2 m/s**.

E14 → ½mv^2 = ½(4m)v'^2 → v' = **v/2** (speed halves).

E15 → **No.** Gravity is conservative (work depends only on the height difference), but friction is non-conservative and always does negative work $-f_k \cdot d$ proportional to path length, converting mechanical energy into heat. So KE + PE is not conserved (total energy including heat still is).

[HARDER]

[H16] A 3 kg block slides down a 30° incline of length 5 m with μ_k = 0.15. Find its speed at the bottom using energy. (answer: work by gravity = mg sinθ × d = 3×9.8×0.5×5 = 73.5 J; N = mg cosθ = 25.46 N; f_k = 3.82 N; work by friction = −3.82×5 = −19.1 J; net = 54.4 J = ½×3×v^2 → v = 6.03 m/s)

[H17] A spring (k = 500 N/m) is compressed 0.2 m and launches a 0.5 kg block on a frictionless surface. Find the launch speed. (answer: ½kx^2 = ½mv^2 → 500×0.04 = 0.5v^2 → v = √40 = 6.32 m/s)

[H18] Derive why the centripetal force does no work, and connect it to why circular motion at constant speed keeps KE constant. (answer: force ⊥ displacement at every instant, so W = ∫F·ds = 0; by the work–energy theorem ΔKE = 0, so speed stays the same.)

BLOCK 7 — CENTRE OF MASS (L9)

CONCEPT 7.1 — What it is

The **centre of mass (CoM)** is the average position of the mass in a system — the single point that moves as if all the mass were concentrated there and all external forces acted there.

CONCEPT 7.2 — The formula

For particles on a line: $\mathbf{x_{cm} = (m_1x_1 + m_2x_2 + \ldots) / (m_1 + m_2 + \ldots)}$.

In 2D, do each axis separately: $x_{cm} = \Sigma m_i x_i / M$ and $y_{cm} = \Sigma m_i y_i / M$, then write the point ($x_{cm}$, y_{cm}).

Vector form: $\mathbf{R_{cm} = \Sigma m_i r_i / M}$.

CONCEPT 7.3 — Symmetry shortcut

For a uniform symmetric body the CoM is at the geometric centre:

- Uniform rod → midpoint.
- Uniform disc, cube, sphere → geometric centre.
- L-shaped object → split into rectangles, treat each piece's CoM as a point mass.

CONCEPT 7.4 — How the CoM moves

The CoM of an isolated system (no net external force) moves with **constant velocity** — internal forces cannot change it. When mass is redistributed (e.g. ice melting, a person walking on a boat) the CoM of the whole system shifts to keep this true.

HIS QUESTIONS

[HIS M1] Two particles: 2 kg at x = 0 and 8 kg at x = 5. Find the CoM.

[HIS M2] Three particles: 2 kg at (0,0), 3 kg at (4,0), 5 kg at (10,0). Find the CoM.

[HIS M3] A 2 kg and an 8 kg mass are 10 m apart. Where is the CoM measured from the 2 kg mass?

[HIS M4] 2 kg at (0,0), 3 kg at (6,0), 5 kg at (0,4). Find the CoM.

[HIS CONTEST-2 — the one he set and nobody solved] An ice cube of mass m and side L melts in a tray of mass M. The water spreads out thin on the tray base. Does the centre of mass move? If so, in which direction, and what determines the size of the change? Explain WITHOUT numerical calculation.

ANSWERS

M1 → x_cm = (2×0 + 8×5)/10 = **4 m**.

M2 → x_cm = (0 + 12 + 50)/10 = **6.2 m**.

M3 → x_cm = (8×10)/10 = **8 m** from the 2 kg mass (closer to the heavier mass — the CoM always sits nearer the larger mass).

M4 → x_cm = (0 + 18 + 0)/10 = **1.8**; y_cm = (0 + 0 + 20)/10 = **2.0** → **(1.8, 2.0)**.

CONTEST-2 → Yes, it moves. While solid, the ice's mass sits at height L/2, so the combined CoM is slightly above the base, about (mL/2)/(M+m). After melting, the water is a thin layer at the base, so its mass is at base level. The mass has been redistributed downward, so the combined CoM **drops toward the base**. The size of the drop is roughly (mL/2)/(M+m): larger when the ice is heavy compared to the tray (m ≫ M) and smaller when the tray is heavy (M ≫ m). Direction is always downward.

Template for any variant of this question: (i) does the CoM move? yes/no (ii) which direction (iii) what sets the magnitude (the mass ratio and the height change).

[HARDER]

[H19] A rod of mass 2 kg and length 4 m has a 1 kg mass attached to one end. Where is the CoM measured from that end? (answer: treat the rod's mass at its own centre, 2 m from the end: x_cm = (1×0 + 2×2)/3 = **1.33 m**)

[H20] An L-shape is made of two uniform rods: one 3 kg × 4 m horizontal, one 4 kg × 3 m vertical joined at one end. Find the CoM. (answer: horizontal piece CoM at (2, 0); vertical piece CoM at (0, 1.5); x_cm = (3×2 + 4×0)/7 = 0.86 m; y_cm = (3×0 + 4×1.5)/7 = 0.86 m → (0.86, 0.86))

[H21] A man of 60 kg walks from one end of a 120 kg boat (length 4 m) to the other. If the boat is free to slide on water, how far does the boat move? (answer: the CoM of the man+boat system does not move, so the boat slides

60/(60+120) × 4 = **1.33 m** opposite to his walk.)

BLOCK 8 — MOMENTUM & COLLISIONS (L10)

CONCEPT 8.1 — Momentum

p = mv (mass × velocity). It is a vector, so it has direction. Units kg·m/s (= N·s).

CONCEPT 8.2 — Impulse

Impulse = force × time = change in momentum: **J = FΔt = Δp = m(v – u)**. This is why a longer collision time (airbag, padding) reduces the peak force for the same change in momentum.

CONCEPT 8.3 — Conservation of linear momentum

If the **net external force on a system is zero**, the total momentum is conserved: **p_before = p_after**.

Internal forces (the collision forces between the two bodies) cancel in pairs (Newton's third law), so they cannot change the total.

This is the single most important equation in this block.

CONCEPT 8.4 — Types of collision

- **Perfectly inelastic (e = 0):** the bodies stick and move with one common velocity $v = (m_1u_1 + m_2u_2)/(m_1+m_2)$. Maximum possible loss of kinetic energy.
- **Elastic (e = 1):** both momentum AND kinetic energy are conserved.
- **Inelastic (0 < e < 1):** momentum conserved, kinetic energy partly lost.

Special theorem: for two EQUAL masses in an elastic collision with the second at rest, they **exchange velocities** (the first stops, the second takes all the speed).

CONCEPT 8.5 — Coefficient of restitution

e = separation speed / approach speed = $(v_2 – v_1)/(u_1 – u_2)$. It measures how "bouncy" a collision is. e = 1 elastic, e = 0 perfectly inelastic.

CONCEPT 8.6 — The two-equation machine (solve every general collision with this)

1. Momentum: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

2. Restitution: $v_2 – v_1 = e(u_1 – u_2)$

Two equations, two unknowns (v_1, v_2). Solve them simultaneously.

CONCEPT 8.7 — Multi-dimensional collisions

Resolve everything into x and y components. Conserve momentum separately on each axis, then combine: $|p| = \sqrt{(p_x^2 + p_y^2)}$, direction $\theta = \tan^{-1}(p_y/p_x)$.

CONCEPT 8.8 — Centre of mass link

If $F_{net} = 0$: total momentum constant, CoM velocity constant, CoM acceleration = 0. And $a_{cm} = F_{net}/M$ (works even when $F_{net} \neq 0$).

HIS QUESTIONS

[HIS P1] 2 kg at 6 m/s hits a stationary 4 kg. What is the total momentum before the collision?

[HIS P2] 5 kg at 4 m/s hits a stationary 3 kg and they stick. Find the common velocity.

[HIS P3] 2 kg at 8 m/s hits a stationary 6 kg perfectly inelastically. Find initial momentum, common velocity, initial KE, final KE, and energy lost.

[HIS P4] 2 kg at 10 m/s elastically hits a stationary 3 kg. Find both final velocities.

[HIS P5] 4 kg at 6 m/s collides head-on with 2 kg at 3 m/s in the opposite direction. Find both final velocities (elastic).

[HIS P6] 5 kg at 8 m/s hits a stationary 3 kg and they stick. Find the common velocity, initial KE, final KE, and energy lost.

[HIS P7] 2 kg at 10 m/s hits a stationary 3 kg with e = 0.6. Find both final velocities.

[HIS P8] 4 kg at 8 m/s hits a stationary 6 kg with e = 0.5. Find both final velocities.

[HIS P9] 3 kg at 6 m/s hits a stationary 2 kg with e = 0.4. Find the approach speed, separation speed, and both final velocities.

[HIS P10] 2 kg at 12 m/s hits a stationary 4 kg with e = 0.5. Find both final velocities.

[HIS P11] 4 kg moving east at 5 m/s sticks to 6 kg moving north at 3 m/s. Find the final speed and direction.

[HIS P12] 5 kg moving along +x at 8 m/s sticks to 3 kg moving along +y at 6 m/s. Find the final speed and direction.

[HIS P13] A 0.5 kg ball moving at 10 m/s hits a wall and rebounds at 6 m/s in the opposite direction. Find initial momentum, final momentum, change in momentum, and the impulse on the ball.

[HIS P14] A 10 kg body at rest explodes into two pieces, 4 kg at 12 m/s and a 6 kg piece. Find the velocity of the second piece.

[HIS P15] 2 kg east at 6 m/s and 3 kg north at 4 m/s collide. Find the components, magnitude, and direction of the total momentum.

[HIS P16] 2 kg east at 5 m/s and 1 kg north at 4 m/s stick together. Find the final speed and direction.

[HIS P17] If the net external force on a system is zero, what can you say about (a) the total momentum (b) the velocity of the CoM (c) the acceleration of the CoM?

ANSWERS

P1 → 2×6 = **12 kg·m/s**.

P2 → v = (5×4)/8 = **2.5 m/s**.

P3 → p_i = 16; v = 16/8 = **2 m/s**; KE_i = 64 J; KE_f = 16 J; lost = **48 J**.

P4 → $v_1 = (m_1 - m_2)u/(m_1+m_2) = (2-3)\times 10/5 =$ **–2 m/s** (rebounds back); $v_2 = 2m_1u/(m_1+m_2) =$ **8 m/s**.

P5 → Take $u_1 = 6$, $u_2 = -3$. $v_1 = [(4-2)\times 6 + 2\times 2\times(-3)]/6 =$ **0**; $v_2 = [(2-4)(-3) + 2\times 4\times 6]/6 =$ **9 m/s**.

P6 → v = 40/8 = **5 m/s**; KE_i = 160 J; KE_f = 100 J; lost = **60 J**.

P7 → $v_2 - v_1 = 6$; $2v_1 + 3v_2 = 20$ → v_1 = **0.4**, v_2 = **6.4 m/s**.

P8 → $v_2 - v_1 = 4$; $4v_1 + 6v_2 = 32$ → v_1 = **0.8**, v_2 = **4.8 m/s**.

P9 → approach = 6, separation = e×6 = **2.4 m/s**; $3v_1 + 2v_2 = 18$ and $v_2 - v_1 = 2.4$ → v_1 = **2.64**, v_2 = **5.04 m/s**.

P10 → $v_2 - v_1 = 6$; $2v_1 + 4v_2 = 24$ → v_1 = **0**, v_2 = **6 m/s** (the lighter body stops dead).

P11 → p_x = 20, p_y = 18, total mass 10 → $v = \sqrt{(400+324)}/10 = \sqrt{724}/10 \approx$ **2.69 m/s**; $\theta = \tan^{-1}(18/20) \approx$ **42° north of east**.

P12 → p_x = 40, p_y = 18, mass 8 → $v \approx$ **5.48 m/s**; $\theta = \tan^{-1}(18/40) \approx$ **24° from the x-axis**.

P13 → $p_i = 5$; $p_f = -3$; $\Delta p =$ **–8 kg·m/s**; impulse = **–8 N·s** (opposite to the original direction).

P14 → $0 = 4\times12 + 6v$ → $v =$ **–8 m/s** (8 m/s in the opposite direction).

P15 → p_x = 12, p_y = 12; |p| = **17.0 kg·m/s**; $\theta =$ **45° north of east**.

P16 → p_x = 10, p_y = 4, mass 3 → $v = \sqrt{(100+16)}/3 \approx$ **3.57 m/s**; $\theta = \tan^{-1}(4/10) \approx$ **21.8° north of east**.

P17 → (a) total momentum is constant/conserved (b) the CoM velocity is constant (c) the CoM acceleration is **zero**.

[HARDER]

[H23] A 3 kg ball at 5 m/s collides elastically with a 1 kg ball at rest. Find both final velocities using the two-equation machine with e = 1. (answer: $v_1 = [(3-1)\times5]/4 = 2.5$ m/s; $v_2 = [2\times3\times5]/4 = 7.5$ m/s)

[H24] In a collision, the initial kinetic energy is 200 J and e = 0.8. What is the maximum possible fraction of KE that can be lost? Use $\Delta KE = \frac{1}{2}m_1m_2(u_1-u_2)^2(1-e^2)/(m_1+m_2)$. (answer: fraction lost = $(1-e^2) = 0.36$, so up to 36% — 72 J)

[H25] A 60 kg person jumps off a 120 kg boat at 3 m/s relative to the boat. How fast does the boat move? (answer: by conservation of momentum $0 = 60\times3 + 120\times v$ → $v =$ **1.5 m/s** opposite to the jump)

THE 9 EXAM-DAY TEMPLATES (his favourite shapes)

1. **Error propagation** → value, then Δ by adding relatives, then relative, then %.
2. **a(t)** → **v(t)** → **x(t)** → integrate twice, pin the constants from the initial conditions, then evaluate.
3. **Velocity–time graph story** → draw the ramp/flat, displacement = area, average velocity = displacement/time, average acceleration = Δv/Δt.
4. **Projectile (a)(b)(c)** → components, then time of flight, then max height (add range if asked).
5. **Collision with e** → momentum equation + restitution equation → solve the system.
6. **2-Dimensional stick** → conserve p_x and p_y separately → $v = \sqrt{(v_x^2 + v_y^2)}$, $\theta = \tan^{-1}(p_y/p_x)$.
7. **Incline + pulley with friction** → whole-system a = (driving – resisting)/total mass → tension from one body.
8. **CoM in 2D** → weighted means per axis.
9. **Ice-tray style explanation** → claim → redistribution → direction → mass-ratio controls the magnitude.

FORMULA CARD (copy this onto one page)

- $v = u + at$; $v^2 = u^2 + 2as$; $s = ut + \frac{1}{2}at^2$

- Projectile: $T = 2u_y/g$, $H = u_y^2/2g$, $R = u^2\sin 2\theta/g$
- Relative velocity: $v(A/B) = v_A - v_B$
- Errors: multiply/divide → add relatives; power n → ×n; add/subtract → add absolutes
- Circular: $\omega = v/r$, $a_c = v^2/r = r\omega^2$, $a_t = r\alpha$, $\tau = rF\sin\theta$
- Friction: $f_k = \mu_k N$, $f_s \le \mu_s N$, incline slides iff $\tan\theta > \mu_s$
- Work/Energy: $W = Fd\cos\theta$, $KE = \frac{1}{2}mv^2$, $PE = mgh$, $W_{net} = \Delta KE$
- Momentum: $p = mv$, $J = F\Delta t = \Delta p$, $p_{before} = p_{after}$
- CoM: $x_{cm} = \Sigma mx/\Sigma m$, $v_{cm} = \Sigma mv/\Sigma m$, $a_{cm} = F_{net}/M$
- Restitution: $e = (v_2 - v_1)/(u_1 - u_2)$

EXAM-DAY PROTOCOL (calibrated to how his papers actually mark)

1. **First pass (35 min):** all five 3-mark questions. These are definitions and one-step calculations from the flash cards and CONCEPT boxes. Never leave one blank — write the formula and one sentence. Partial marks exist.

2. **Second pass (60 min):** the three 5-mark questions. Use the (A)/(B) choice to pick the pair whose technique you can write from memory. Label the sub-parts (a)(b)(c). Show numbers and units on every line.

3. **Last 15 min:** the 10-mark story question. Use the four-verb structure: (a) evaluate the claim (b) support with calculation (c) design/decide (d) assess the limitation. Four labelled paragraphs beat one long one.

4. **If a question looks new:** it is one of his worksheets with different numbers. Identify the block, pull the matching template, re-run the numbers.

5. **g:** read it from the paper. Worksheets use 9.8, one contest used 10. sin37 = 0.6, cos37 = 0.8 when needed.

6. **Three-minute rule:** stuck more than 3 min? Write the formula and what you know, move on, return at the end.

7. **Marks live in the method.** Show every step, add units, and use words like "because", "therefore", "since the net force is zero". His papers reward written justification over bare numbers.

All the best tomorrow.